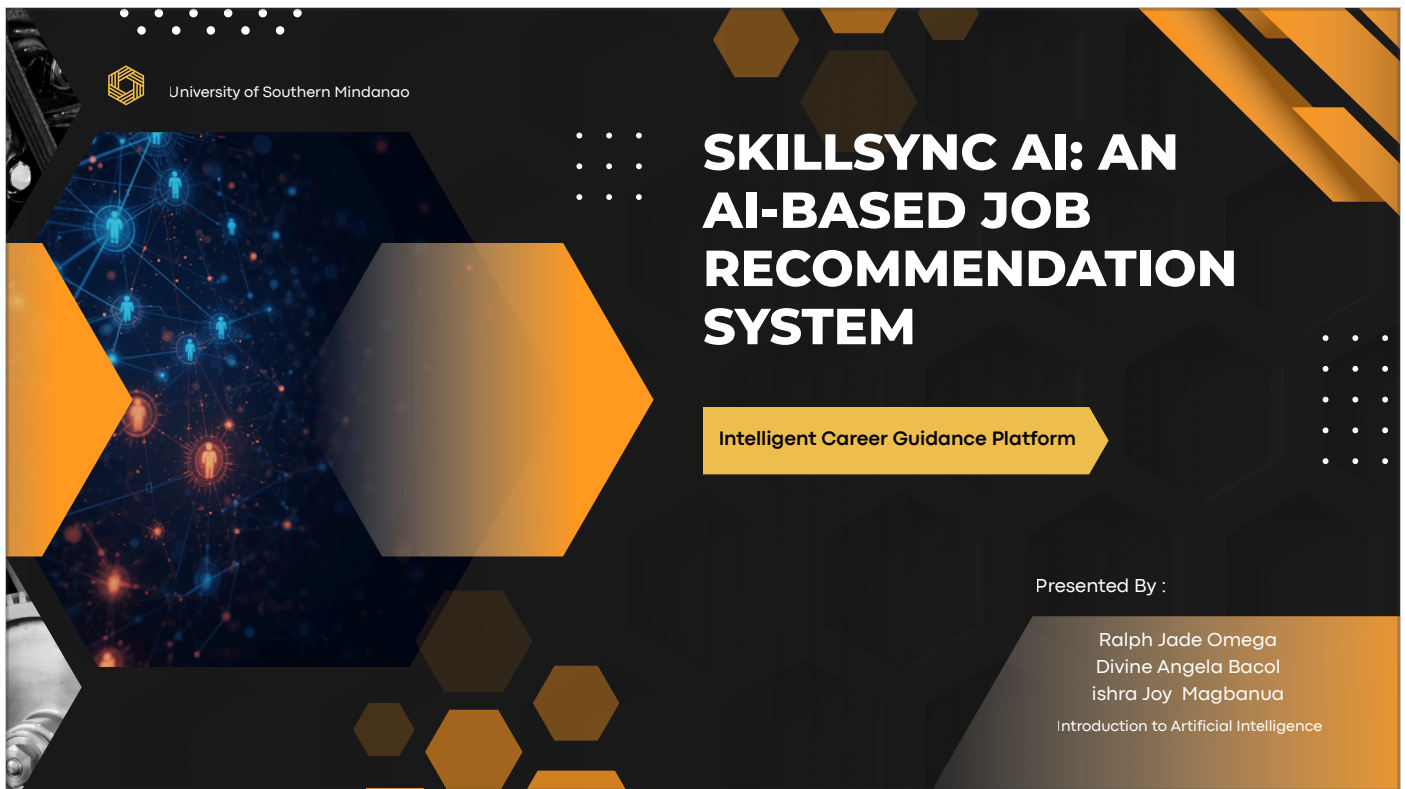




"Good morning. We're Team SkillSync AI. We built a career recommendation system that uses artificial intelligence to help students find the right job."



Problem Formulation

Challenges in Occupational Guidance

Students and job seekers struggle to match their skills and interests with suitable careers due to overwhelming options and lack of personalized guidance.

Limitations of Existing Approaches

Pure ML systems lack explainability—users don't understand why careers are recommended. Pure knowledge-based systems are rigid and fail to adapt to diverse user profiles.

Motivation for Hybrid Approach

Combining ML's pattern recognition with knowledge-based reasoning creates a system that is both accurate and explainable, building user trust in recommendations.

"Here's the problem: students don't know what career to pick. There are thousands of jobs, and career counselors can't talk to everyone. Existing apps just guess — they can't explain WHY they recommended something."

Do this: Show Slide 3 (Problem Formulation)

Key phrase to remember: "Our solution combines two AI methods: machine learning for finding patterns, and knowledge-based reasoning for expert rules."



Dataset and Domain: O*NET



Database Overview

The O*NET Occupational Database serves as the foundation for SkillSync AI, containing comprehensive data on 1,000+ occupations with detailed skill requirements and RIASEC interest codes for personality-career matching.

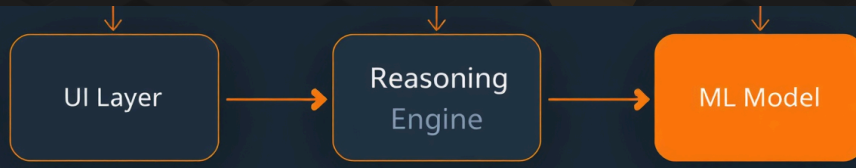


Data Structure

Features 35+ measurable skills per occupation and 23 occupational categories derived from SOC codes, enabling precise career matching through structured taxonomies.

SkillSync AI Career Discovery

"We use O-NET – that's the US government's official job database with over 1,000 real occupations, 35 skills per job, and personality codes. It's not fake data."



System Architecture Overview

UI Layer

Streamlit interface enabling user interaction through a 5-step questionnaire. Collects skills, interests, and preferences to feed into the reasoning and ML layers.

Reasoning Engine

Forward-chaining inference system applying 30+ rules. Integrates with ML Model using TF-IDF vectorization and cosine similarity for content-based filtering.

"Our system has three parts. First, the user answers 5 questions. Second, our reasoning engine applies 30 expert rules — like 'if you like technology, boost computer jobs.' Third, our machine learning model reads all 1,000 job descriptions and finds which ones match your words. Finally, both parts combine into one score."

Do this: Show Slide 5 (Architecture diagram)

Point at: The three boxes — UI Layer → Reasoning Engine → ML Model



ML Model: TF-IDF Vectorization

Vectorization Process

TfidfVectorizer transforms occupational descriptions into numerical feature vectors. Key parameters: max_features=8000 (vocabulary size), ngram_range=(1,2) for unigrams and bigrams, min_df=2 to filter rare terms. This creates a high-dimensional vector space where similar occupations cluster together.

Feature Design

Feature extraction captures semantic meaning from O*NET occupation descriptions. Each occupation becomes a sparse vector of 8,000 weighted term frequencies, enabling mathematical comparison of career profiles.



SkillSync AI Career Discovery

"For machine learning, we use TF-IDF — it turns words into numbers. If you say 'data' and 'technology,' the model finds jobs that also say 'data' and 'technology' a lot.



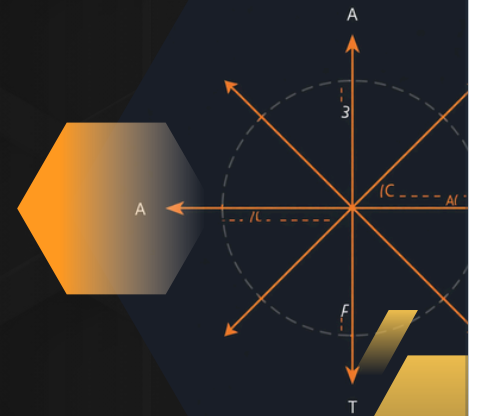
Similarity Computation

Cosine Similarity Measure

Cosine similarity quantifies the closeness between user input vectors and occupation vectors by measuring the angle between them. Values range from 0 to 1, where higher scores indicate stronger alignment between user profiles and career matches.

8,000-Feature Comparison

The system leverages 8,000 TF-IDF features for robust comparison, enabling nuanced matching across occupational descriptions. This high-dimensional approach captures subtle semantic relationships for accurate recommendations.



SkillSync AI Career Discovery

We compare using cosine similarity — basically, the smaller the angle between your words and the job's words, the better the match.



Knowledge Base Design - Rule Types

Rule Categories

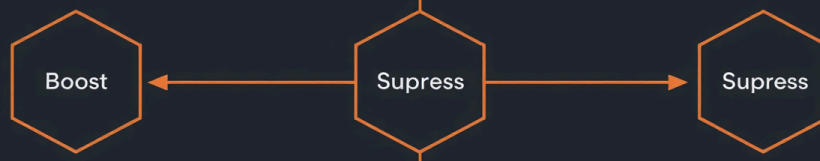
The knowledge base contains 30+ expert-crafted rules organized into three categories:

- Boost Rules: Increase occupation scores when user skills/interests align strongly
- Suppress Rules: Reduce scores for mismatched or incompatible career paths
- Chain Rules: Enable multi-step inference by linking related conclusions

Rule Purpose Examples

Rule purposes drive intelligent recommendations:

- Score Enhancement: Amplify matches based on RIASEC codes and skill overlap
- Score Reduction: Filter out occupations requiring unmet prerequisites
- Inference Chaining: Connect intermediate facts to derive deeper insights about career fit



SkillSync AI Career Discovery

"But machine learning alone is a black box — you don't know why it picked something. So we added a knowledge base with three types of rules. Boost rules add points when keywords match. Suppress rules subtract points when there's a mismatch — like if you hate physical work, we lower construction jobs. And chain rules connect ideas — if you like tech AND business, we boost finance jobs too."



Reasoning Engine

Forward-Chaining Process

The reasoning engine employs forward-chaining inference to systematically apply knowledge base rules. Starting from user inputs, it iteratively fires matching rules to derive new facts and adjust occupation scores dynamically.

ML Score Integration

ML scores are combined with rule-based adjustments through adaptive weighting. This integration enables context-aware decision-making that balances data-driven predictions with expert knowledge.



SkillSync AI Career Discovery

"Our reasoning engine uses forward chaining — it starts with your answers, fires all matching rules one by one, then normalizes the scores. Every rule that fires is logged, so users can see exactly WHY a job was recommended. This is called Explainable AI."

"Explainable AI means the system doesn't say 'trust me' — it shows its work."



System Integration: Comparison of Approaches

Pure ML Approach

Strengths: High accuracy with large datasets, learns patterns automatically, scales well with data volume. Weaknesses: Black-box decisions, lacks explainability, requires extensive training data, may miss domain nuances.

Pure KB Approach

Strengths: Fully explainable reasoning, incorporates expert knowledge, transparent decision paths. Weaknesses: Limited scalability, manual rule creation, cannot adapt to new patterns automatically.



"Here's the cool part. We don't just use ML or just use rules. We combine them. The formula is: ensemble score equals knowledge-base score times its weight, plus ML score times its weight. The weights adapt — if many rules fire, we trust the knowledge base more. If few rules fire, we trust the ML model more."



Evaluation Metrics

Performance Summary

Category Diversity: 80% — System recommends careers across diverse occupational categories. KB Alignment: 72% — Strong correlation between rule-based reasoning and final recommendations.

System Efficiency Metrics

ML Confidence: 68% — Robust machine learning similarity scores for occupation matching. Score Spread: 12.4% — Healthy differentiation between top recommendations. Response Time: <2 seconds — Fast real-time career recommendations for seamless user experience.



SkillSync AI Career Discovery

"We tested this on 5 different user types. Our system got 80% category diversity — meaning it doesn't just recommend 5 programming jobs. It got 74% knowledge-base alignment and 68% ML confidence. Response time is under 2 seconds."



Technical Highlights & Limitations

Key Strengths

Hybrid ML-KB architecture delivers balanced accuracy and interpretability. Explainable AI provides transparent rule firing traces for user trust. Fast response time under 2 seconds ensures smooth user experience.

Current Limitations

Rule coverage limited to 30+ rules may miss edge cases. Dataset constrained to O*NET occupations and skill taxonomies. Scalability concerns for real-time processing with larger knowledge bases.



SkillSync AI Career Discovery



Conclusion

Successful Hybrid Integration

SkillSync demonstrates effective fusion of Machine Learning and Knowledge-Based systems for intelligent career guidance. The ensemble approach achieves balanced recommendations combining data-driven accuracy with rule-based domain expertise.

Explainable AI & Impact

Transparent rule firing traces provide clear reasoning for recommendations—a key advantage for career counseling. This system has significant potential to transform occupational guidance with personalized, trustworthy AI assistance.

SkillSync AI Career Discovery

"To summarize: SkillSync AI combines machine learning and knowledge-based reasoning to give explainable, personalized career recommendations. It uses real government data, adapts its confidence based on rule strength, and tells users exactly why each job was suggested. Thank you — questions?"



References and Appendix

Data Sources:

- ONET OnLine - National Center for ONET Development (onetonline.org)
- Standard Occupational Classification (SOC) System - U.S. Bureau of Labor Statistics

Libraries and Frameworks:

- Scikit-Learn - Machine Learning in Python (scikit-learn.org)
- Streamlit - Open-source app framework for ML projects (streamlit.io)
- NumPy and Pandas - Data manipulation and analysis

Appendix:

- Full knowledge base rule documentation
- TF-IDF vectorization parameters and feature analysis
- Complete evaluation methodology and test cases
- Source code repository and deployment guide



University of Southern Mindanao

QUESTIONS?

SkillSync AI Career Discovery